

The University of Zambia
Department of Mathematics and Statistics
Mat 1110 - Foundation Mathematics and Statistics for
Social Sciences

Tutorial Sheet 6 - Inequalities & Modulus Functions

July, 2023.

1. Solve each of the following quadratic inequalities, in each case, illustrating the solution set on the number line.

(a) $2x^2 + 5x \geq 3$	(d) $x^2 - 3x - 4 \neq 0$	(g) $x^2 - 2x + 1 < 0$
(b) $x^2 - 21x + 110 < 0$	(e) $x^2 + 2x + 9 < 0$	(h) $x^2 + 9 \leq 0$
(c) $7x^2 - 29x \leq -4$	(f) $-2x + 7 \geq -5x$	(i) $-x^2 < 0$

2. Sketch each of the following quadratic functions. In each case, state the set of value(s) of x for which $f(x) > 0$, $f(x) \leq 0$.

(a) $f(x) = -2x^2 - 5x + 3$	(c) $f(x) = 2x^2 - 4x + 5$	(e) $f(x) = x^2 - 1$
(b) $f(x) = -x^2 - 4x + 1$	(d) $f(x) = -2x^2 - 3x + 2$	(f) $f(x) = x^2 + 5x + 6$

3. Solve each of the following cubic inequalities, in each case, illustrating the solution set on the number line.

(a) $4x^3 - x < 6 - 9x^2$	(c) $x^3 - x^2 - 9x + 9 \leq 0$
(b) $(x + 4)(x - 2)(x - 7) > 0$	(d) $4x^3 - 8x^2 - x + 2 \neq 0$

4. Sketch each of the following cubic functions. In each case, state the set of value(s) of x for which $f(x) > 0$, $f(x) \leq 0$.

(a) $f(x) = 4x^3 + 9x^2 - x - 6$	(c) $f(x) = x^3 - 5x^2 + 3x + 9$
(b) $f(x) = x^3 - 5x^2 - 22x + 56$	(d) $f(x) = x^4 + 4x^3 - 12x^2$

5. Solve each of the following rational inequalities, in each case, illustrating the solution set on the number line.

$$\begin{array}{lll}
 \text{(a)} \quad \frac{x-1}{x+2} \leq 7 & \text{(d)} \quad \frac{4}{x^2-4x-5} \leq 0 & \text{(g)} \quad \frac{x-3}{x+4} \geq \frac{x+2}{x-5} \\
 \text{(b)} \quad \frac{x+2}{x-1} \geq \frac{-1}{2} & \text{(e)} \quad \frac{x}{x+1} \leq \frac{x}{x+2} & \text{(h)} \quad \frac{2}{x^2+4} \geq 0 \\
 \text{(c)} \quad \frac{x^2-9}{x^2+1} < 0 & \text{(f)} \quad x(x-1) > \frac{x-1}{x} & \text{(i)} \quad \frac{x^2-2x+2}{(x-1)^2+7} > 0
 \end{array}$$

6. Sketch the graph of each of the following functions

$$\begin{array}{lll}
 \text{(a)} \quad f(x) = |x| - 2 & \text{(d)} \quad f(x) = |x + 7| & \text{(g)} \quad f(x) = |x^3 + 2x^2 - 3x| \\
 \text{(b)} \quad f(x) = |x| + 1 & \text{(e)} \quad f(x) = |x^2 + 5x + 6| & \\
 \text{(c)} \quad f(x) = |x - 3| & \text{(f)} \quad f(x) = |-x^2 + 1| &
 \end{array}$$

7. On the same graph, sketch each of the following functions

$$\begin{array}{ll}
 \text{(a)} \quad f(x) = |x - 1| \text{ and } f(x) = |x + 2| & \text{(b)} \quad f(x) = |x| \text{ and } f(x) = |x^2 - 2|
 \end{array}$$

8. Find all values of x that satisfy each of the following equations

$$\begin{array}{lll}
 \text{(a)} \quad |x + 1| = -3 & \text{(d)} \quad |x| = 2x + 10 & \text{(g)} \quad |x + 5| = |x + 1| \\
 \text{(b)} \quad 2x = |-3x + 8| & \text{(e)} \quad |3x - 7| = |x + 5| & \text{(h)} \quad |-x + 2| = 5 \\
 \text{(c)} \quad |x + 1| = -2x - 5 & \text{(f)} \quad |3x + 9| = |2x + 1| & \text{(i)} \quad |-3 - 2x| = -x
 \end{array}$$